THE CROSS-ALIGNMENT MATRIX

ALEXANDER S. PETTY

Abstract. We introduce the cross-alignment matrix of a fractional field: the symmetric matrix $\mathbf{A}(n)$ whose (i, j) entry is the position-wise digit-match proportion between i/n and j/n. For digit-partitioning primes, $\mathbf{A}(p) = I$. For $n = ps$ with p digit-partitioning and s smooth, $\mathbf{A}$ decomposes into p blocks of rank 1, giving $\operatorname{rank}(\mathbf{A}) = p$.

The pairwise alignment $\sigma(n)$ is the mean off-diagonal entry. For primitive-root primes, the matrix is permutation-similar to a circulant whose eigenvalues are the DFT of the cyclic autocorrelation.

1. Introduction

Scalars measure one thing at a time. The matrix measures everything at once.

The preceding papers measured the fractional field $\mathcal{F}(n) = \{k/n : 1 \leq k \leq n-1\}$ through $\alpha(n)$, $\sigma(n)$, and $F(n)$. This paper introduces the object they are projections of: the cross-alignment matrix.

2. Definition

Definition 1. The *cross-alignment matrix* of n in base b is the $(n-1) \times (n-1)$ symmetric matrix $\mathbf{A}(n)$ with entries

$$A_{ij} = a(i/n,\, j/n),$$

where $a(i/n,\, j/n)$ is the position-wise digit-match proportion between the repetends of i/n and j/n. Terminating pairs contribute 1; pairs with different cycle lengths contribute 0; the diagonal satisfies $A_{ii} = 1$.

The matrix is symmetric by construction ($A_{ij} = A_{ji}$) and positive semidefinite. To see this, place each fraction into a direct-sum feature space indexed by its comparison class. For a terminating fraction, use the same unit vector e_{term}. For a non-terminating fraction with period length L, use the normalized one-hot vector in $\mathbb{R}^{L \times b}$ whose (ℓ, d) entry is 1 exactly when the ℓ-th digit of its repetend equals d. Two fractions with different cycle lengths lie in orthogonal summands, so their inner

Date: April 2021.

2020 *Mathematics Subject Classification.* 11A63, 11B83, 42A16.

product is 0; two terminating fractions have inner product 1; and two fractions with the same period length have inner product equal to the proportion of matching digits at aligned positions. Thus $\mathbf{A}(n)$ is a Gram matrix and is therefore PSD [2].

3. Recovery of Scalar Invariants

The pairwise alignment σ is recovered directly from $\mathbf{A}$:

$$\sigma(n) = \binom{n-1}{2}^{-1} \sum_{i<j} A_{ij} \qquad \text{(mean off-diagonal entry).}$$

The alignment $\alpha(n)$ is not determined by $\mathbf{A}$ alone, because α counts terminating fractions as aligned with $1/n$ (contributing 1), while $\mathbf{A}$ assigns 0 to pairs with different cycle lengths. The matrix captures the non-terminating pairwise structure. The alignment α additionally requires the terminating-fraction convention from the earlier papers.

4. Spectral Structure for Digit-Partitioning Primes

Theorem 2. *If p is a digit-partitioning prime in base b (i.e., $p \leq b+1$), then $\mathbf{A}(p) = \mathbf{I}_{p-1}$, the identity matrix.*

Proof. By the digit-partitioning property, no two distinct fractions i/p and j/p share a digit at any common position. Thus $A_{ij} = 0$ for $i \neq j$ and $A_{ii} = 1$. □

All eigenvalues equal 1. The field is maximally non-degenerate: every fraction is structurally independent from every other.

5. Spectral Structure for Non-DP Primes

Observation 3. For a prime p not dividing b with $p > b+1$, the eigenvalue spectrum of $\mathbf{A}(p)$ reflects the coset structure of $\langle b \rangle$ in $(\mathbb{Z}/p\mathbb{Z})^*$.

When b is a primitive root modulo p, the matrix is permutation-similar to a circulant [1] (reordering rows and columns by discrete logarithm), with eigenvalues given by the DFT of the cyclic autocorrelation.

When there are exactly two cosets ($L = (p-1)/2$), computation shows the spectrum has two eigenvalue levels whose values depend on the cross-coset digit-match count.

In base 10: $p = 13$ ($L = 6$, two cosets) has eigenvalues $4/3$ and $2/3$, each with multiplicity 6. The prime $p = 17$ ($L = 16$, primitive root) has nine distinct eigenvalues ranging from 0.25 to 1.75.

6. Block Structure

Proposition 4. *For $n = ps$ with p a digit-partitioning prime and s a b-smooth integer, the cross-alignment matrix is permutation-similar to*

$$\mathbf{A}(ps) \cong J_{s-1} \oplus \underbrace{J_s \oplus \cdots \oplus J_s}_{p-1},$$

where J_m is the $m \times m$ all-ones matrix. In particular, $\operatorname{rank}(\mathbf{A}(ps)) = p$.

Proof. The $s-1$ terminating fractions form one block (pairwise alignment 1). The $(p-1)$ exact repetend classes, each of size s, form the remaining blocks (within-class alignment 1, cross-class 0 by the digit-partitioning property). Each all-ones block J_m has rank 1. □

For digit-partitioning $n = ps$, the rank is always p, regardless of s. The null space grows with the smooth factor because each repetend class of size s contributes rank 1 (all members are structurally identical in alignment space).

7. Remarks

What the matrix captures. The pairwise alignment σ is directly recoverable from $\mathbf{A}$. The alignment α requires the additional terminating-fraction convention. For digit-partitioning primes, $\mathbf{A}(p) = I$ characterizes the property completely.

Open directions. Whether the eigenvalue spectrum of $\mathbf{A}(n)$ admits a closed-form description in terms of the prime factorization of n and the digit function is open. The connection to Dirichlet characters [3], suggested by the coset-level structure of the spectrum, is pursued in the papers that follow.

References

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[2] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.
[3] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, Amer. Math. Soc., 2004.

Email address: `alexander.petty@gmail.com`